ATLAS OS

Business Plan

Business & fundraising

> **Autonomous, self-sufficient habitats for everyone — operated by software, funded by humanity.**

Program	ATLAS Habitat Initiative
First unit	ATLAS 01
Operating software	ATLAS OS (the Habitat Twin)
Author	Elijah Royaie
Status	Draft v0.1 — for review
Launch market	Canada !’ United States !’ global
Companion docs	Legal & Compliance · Budget & Fundraising · Pitch Deck · PRD

1. Executive summary

ATLAS designs, builds, and operates **self-sufficient residential buildings** where every floor is dedicated to one human need — water, energy, food, shelter, air, health, restoration. The building generates its own power, recycles its own water and waste, grows a meaningful share of its own fresh food, and is run by **ATLAS OS**, an AI operations layer that sits on top of safety-rated building controls. Construction is modular and partly 3D-printed to drive cost down and enable replication.

The central thesis is a cost-curve inversion: self-sufficiency is _capex-heavy and opex-light_. It costs more to build but almost nothing to run. That inversion is the business model — philanthropy, CSR/ESG capital, impact grants, carbon credits, and SR&ED R&D tax credits fund the upfront hump; residents then live at near-zero ongoing cost, with dignity and independence from fragile external utilities.

We are not selling a cheap building. We are selling **homes people can afford to *keep*** — and, increasingly, the only kind of housing that stays livable when grids, water systems, and supply chains fail.

2. The problem

- **Recurring costs make housing structurally unaffordable.** Utilities + water + food are a perpetual tax on the most vulnerable. You can subsidize a mortgage once; you cannot subsidize a hydro bill forever.
- **Centralized systems are fragile.** Grids, water mains, and food supply chains fail during disasters, conflict, and political instability — exactly when vulnerable people need them most.
- **Humanitarian housing depends on continuous provisioning.** Fuel, food drops, and water trucking are expensive and interruptible.
- **"Smart" buildings get unplugged.** They need skilled operators a property manager doesn't have, so the intelligence is disabled within a year.

3. The solution

A habitat that **provisions itself and runs itself**, legibly enough that a non-engineer can operate it:

- **Energy:** solar PV + battery storage + biogas generator !’ net energy independence, grid as backup only.
- **Water:** rainwater + well capture, multi-stage filtration, ~90%+ greywater reuse, blackwater treatment.
- **Waste !’ resources:** an anaerobic digester turns organic waste into biogas (energy) and liquid fertilizer (feeds the farm).
- **Food:** aquaponics, vertical hydroponics, mushrooms, insect protein, and isolated micro-livestock (eggs). Defined honestly as a **fresh-produce + protein supplement (30–60% of nutritional value), not 100% of calories.**
- **Software:** ATLAS OS — a 3D digital twin + multi-agent AI ops layer that forecasts, optimizes, and triages, while deterministic controls and humans hold life safety.

4. The product line

Layer	What it is	Why it matters commercially
The Habitat	A partly-3D-printed, modular, self-sufficient mid-rise (40–60 units; configurable, incl. up to 12 penthouses + shared rooftop pool)	The physical asset funders endow and residents live in
ATLAS OS	Web-based building-operations platform + AI ops layer	Makes the building operable by a non-technical manager; the replicable, licensable IP
The Kit	A repeatable BOM + 3D-print pipeline + floor "kits"	Turns one building into a hundred; the scaling engine

5. Market & beachhead

- **Beachhead — Canada.** Strong SR&ED R&D incentives, green-building grants, an acute affordable-housing crisis, and clear (if demanding) regulatory regimes. We validate economics and the R&D-funding edge here.
- **Expansion — United States.** Larger affordable-housing and disaster-recovery markets; CSR/ESG capital depth.
- **Global / humanitarian.** Refugee resettlement, remote communities, disaster recovery — where self-provisioning isn't a premium, it's the entire point.

Serviceable segments: affordable & supportive housing operators, Indigenous & remote communities, disaster-recovery agencies, humanitarian/NGO resettlement, and ESG-motivated corporate/philanthropic developers.

6. Business model & revenue

ATLAS is a **blended-capital** venture: philanthropic/impact capital funds the asset; a software + kit business captures recurring, scalable revenue. Pick the lever per deployment — **let the business model drive the tech, not the reverse.**

Stream	Type	Notes
Philanthropy / grants	Capital (capex)	Funds the build hump; humanitarian & foundation grants

CSR / ESG corporate capital	Capital (capex)	Branded floors/buildings + impact reporting
SR&ED tax credits	Non-dilutive	For AI ops, food-loop, anomaly-detection R&D
Carbon credits / green incentives	Recurring/one-time	Emissions + energy avoided
ATLAS OS licensing / SaaS	Recurring (opex play)	Per-building operations platform; the durable margin
Kit / EPC fees	Per-project	Design + 3D-print pipeline + BOM for each new build
Rent / operations	Recurring	Near-zero opex can be a rent premium, an operator opex play, or a retention tool
Crowdfunding / B-Corp equity	Capital	Community ownership and trust

Why charity capital works here (and usually doesn't for housing): a donor's dollar into ATLAS is not consumed — it's *endowed into an asset that keeps producing*. One upfront gift removes a family's utility, water, and partial food costs for 25–30 years. That is an unusually high-leverage philanthropic outcome, and it's measurable (autonomy %, residents served, carbon avoided — see ATLAS OS's impact dashboard, F12).

7. Unit economics (illustrative, CAD)

Order-of-magnitude only; replaced by the parametric model. ~40–60 unit mid-rise.

Bucket	Range
Shell/structure (3D-print + foundation + roof)	~$150–250 / sq ft
Dwelling fit-out	~$50–150 / sq ft
Energy (solar + storage + biogas)	~$0.3–1.5M
Water/waste recycling	~$0.2–0.8M
Food production floors	~$0.5–2M+
Automation / ATLAS OS	~$0.1–0.5M
Skydeck/reservoir	~$0.2–0.6M
All-in (typical)	~CAD $10–30M ("H $200k–500k / unit upfront)

Lifetime framing: high capex !’ near-zero opex. Donated/humanitarian land can cut 20–40%. The north-star metric is **lifetime cost per resident over 25–30 years**, where ATLAS wins decisively against conventional affordable housing carrying perpetual utility and food costs. Full model in Budget & Fundraising.

8. Go-to-market & phasing

Phase	Goal	Commercial milestone
P0 — Concept (now)	PRD, cost model, prototype twin, partner outreach	ATLAS OS prototype live '

P1 — Pilot	Small retrofit/pilot in Canada; validate subsystems + SR&ED R&D	First grant + SR&ED claim; LOIs from a housing operator
P2 — ATLAS 01	First purpose-built habitat, Canada	Anchor philanthropic/CSR funder; residents housed
P3 — Kit	Replicable 3D-print pipeline + repeatable BOM	First licensed/replicated build; OS SaaS contracts
P4 — Expansion	US + international (incl. humanitarian)	Multi-building fleet; data-residency per region

9. Competitive position & moat

- **Open-standards, local-first software** (BACnet/Modbus/Matter + Brick/Haystack) avoids vendor lock-in that kills "smart building" projects.
- **The OS + Kit is the moat**, not any single building: replicability, a tagged telemetry schema enabling fleet-wide AI, and an SR&ED-funded R&D flywheel competitors lack.
- **Honesty as positioning:** we state the hard truths (fresh food not all calories; operational not legal independence; AI never holds life safety). That credibility is what unlocks serious institutional and philanthropic capital.

10. Team & operating model

- **Founder:** Elijah Royaie — AI/multi-agent systems, telehealth/triage, compliance.
- **Needs (to staff per phase):** building-systems/MEP engineering, modular/3D-print construction partner, regulatory/legal counsel (CA + target regions), impact/grants lead, food-systems lead.
- **Structure options** under evaluation (nonprofit, B-Corp, or hybrid foundation + operating co.) — see Legal & Compliance §Structure.

11. Key risks (summary)

Risk	Mitigation
Capex too high to fund	Lead with lifetime-cost thesis; blended funding; donated land; phased pilots
Over-complex to operate	"Operable by a non-technical manager" principle; ruthless simplification
Regulatory / zoning friction	Engage authorities early; frame as resilience, not sovereignty
Food economics overpromised	Define food as a supplement; communicate honestly
AI failure affecting safety	AI never holds life safety; deterministic + human-in-loop
Funder concentration	Blend philanthropy + CSR + non-dilutive + recurring SaaS

Full register in the [PRD §16] and Legal & Compliance.

12. The ask

We are raising a **blended P1 round** to fund a Canadian pilot, the first SR&ED R&D cycle, and the ATLAS OS build-out. See Budget & Fundraising for the use of funds and the Pitch Deck for the investor/donor narrative.

ATLAS is operationally independent, not legally exempt; it supplies fresh food, not all calories; and its AI optimizes but never overrides safety. These constraints are the design, not its limits.